DO YOU KNOW?

Plastic sheets are used in agriculture to grow vegetables. At the end of the season, these plastic sheets are ploughed back into the soil. The plastic sheets break into tiny pieces and get eaten by earth worms, which is harmful to their health and that of soil.

d. Agricultural activities

Fertilizers, pesticides and insecticides used in agriculture can dissolve in rainwater and flow into water bodies such as rivers and lakes. This causes an excess of nutrients such as nitrates and phosphates as well as toxic chemicals in water bodies. It is called Eutrophication. These sunstances can also be harmful to aquatic life.



Figure 13.12 Agricultural waste

e. Industrial waste

Many industries release toxic wastes such as lead, mercury, cyanides, cadmium, etc. If this waste is unregulated and is released into water bodies it has huge impact on humans, plants, animals and aquatic life.



Figure 13.13 Waste water from industries

f. Oil spills

There are large crude oil and natural gas reserves below the sea bed. With the increasing exploration of crude oil in the oceans, accidents in drilling and transporting oil have also increased. Oil spills cause water pollution which is harmful to aquatic life. The oil which remains floating on the water surface blocks sunshine, reduces the dissolved oxygen in water and suffocates marine organisms.



Figure 13.14 Oil spills

g. Thermal pollution

Large amount of water is used for cooling purposes in thermal and nuclear power plants and many industries. Water used for cooling purposes is discharged back to a river or to original water source at a raised temperature and sometimes with chemicals. This causes rise in temperature and decreases the amount of oxygen dissolved in water, which adversely affects the aquatic life.

13.5.3 Common pollutants

Pollutants are generally classified as domestic pollutants, agricultural pollutants and industrial pollutants. The sources and effects of various water pollutants are shown below in Table 13.1.

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